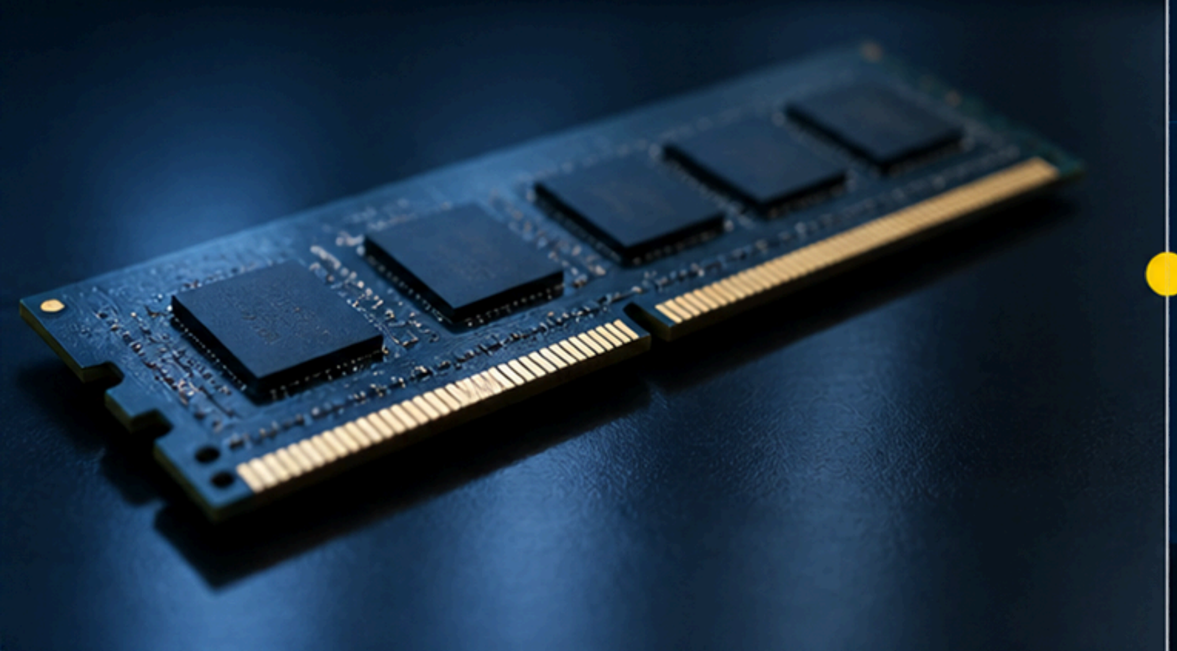
Python-Based Page Replacement Algorithm Simulator

Demonstrating FIFO, LRU, and Optimal Algorithms in Operating Systems

Page Replacement Algorithms Overview



FIFO

Replaces the oldest page in memory.

Simple but may show Belady's anomaly.

LRU

Replaces the page not used for the longest time.

Requires tracking page usage history.

OPTIMAL

Replaces the page not needed for the longest future time.

Minimizes page faults but needs future knowledge.

Simulator Workflow

1. **Provide inputs**
   Enter the page reference string and frame capacity.

2. **Run algorithms**
   Execute FIFO, LRU, and Optimal independently.

3. **Process each reference**
   Check hit or fault and update frames using the algorithm’s replacement policy.

4. **Record outcomes**
   Track page hits and faults throughout the run.

5. **Show results**
   Display step-by-step frames and performance metrics with hit and fault ratios.

Performance Calculations and Formulas

Hit Ratio $= \frac{H}{N}$

Measures the proportion of successful page accesses.

Fault Ratio $= \frac{F}{N}$

Measures the proportion of page faults.

H + F = N

The total of hits and faults equals the total references.

These metrics help measure the efficiency and effectiveness of page replacement.

Sample Results Comparison

Reference string: 1 2 3 1 4 2 2 5 **Frames:** 3 **Total references:** 7

FIFO	LRU	OPTIMAL
Hits: 2	**Hits:** 1	**Hits:** 2
Faults: 5	Faults: 6	Faults: 5
Hit Ratio: 28.57%	Hit Ratio: 14.29%	Hit Ratio: 28.57%
Fault Ratio: 71.43%	Fault Ratio: 85.71%	Fault Ratio: 71.43%

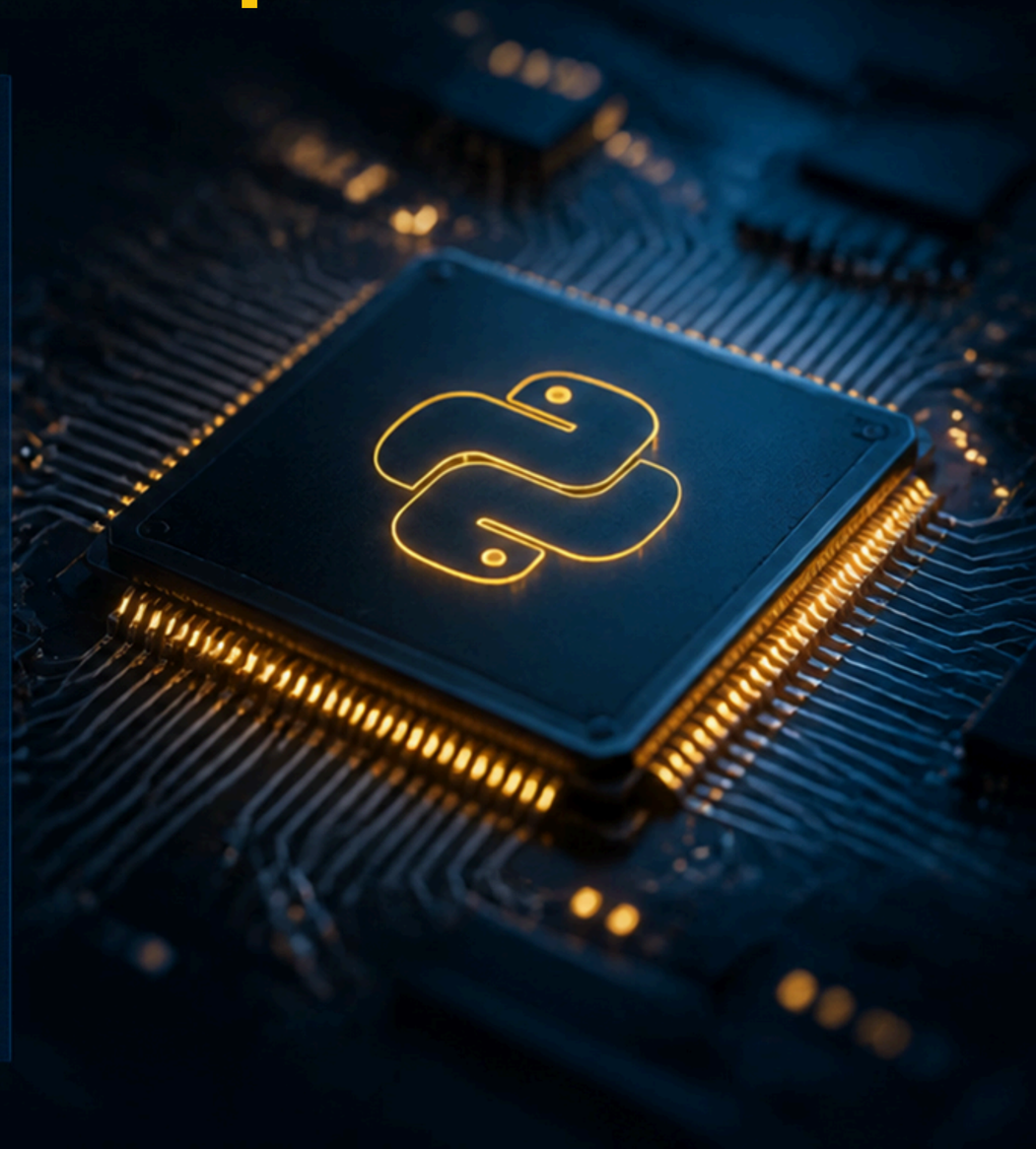
Applications and Future Scope

APPLICATIONS

- Memory management in OS
- Virtual memory systems
- Multitasking and process scheduling
- Performance optimization in computing
- Educational tool for learning OS concepts

FUTURE SCOPE

- GUI development for better interaction
- Addition of algorithms: LFU, Second Chance
- Real-time visualization and charts
- Web-based simulator version
- Enhanced input validation and reports



Conclusion and References

SUMMARY

Simulator bridges OS theory and Python coding.

Enables comparison of key page replacement algorithms.

Helps reduce page faults by informed algorithm choice.

REFERENCES

Operating Systems Lab Notes

Operating System Concepts by Abraham Silberschatz

Python Documentation

Questions and further exploration encouraged.